

Complete **Windows** & **Linux** VM Migration

The Definitive hyper2kvm Feature Guide

Every migration capability in one deck: VirtIO driver injection, initramfs rebuild, registry fixes, BSOD prevention, GRUB repair, SELinux relabeling, fstab UUID repair, boot verification, KubeVirt deployment, batch migration, interactive TUI, and more. From VMware, Hyper-V, AWS, Azure, GCP to KVM — every OS, every source, every workload.

15+

Linux Distros

7

Windows Versions

10

Source Platforms

20+

Automatic Fixes

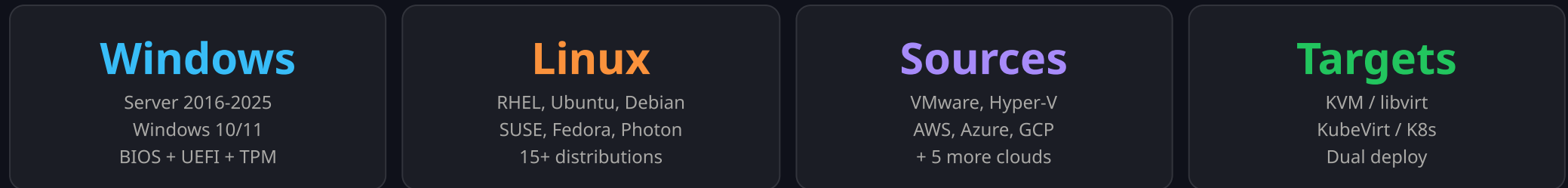
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License Cost

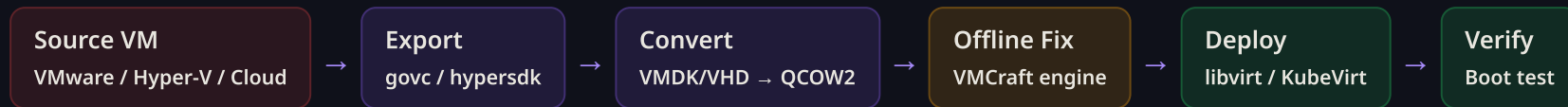
Apache 2.0 | VMCraft Engine | No libguestfs | CLI + TUI + API | KVM + KubeVirt

Platform Overview

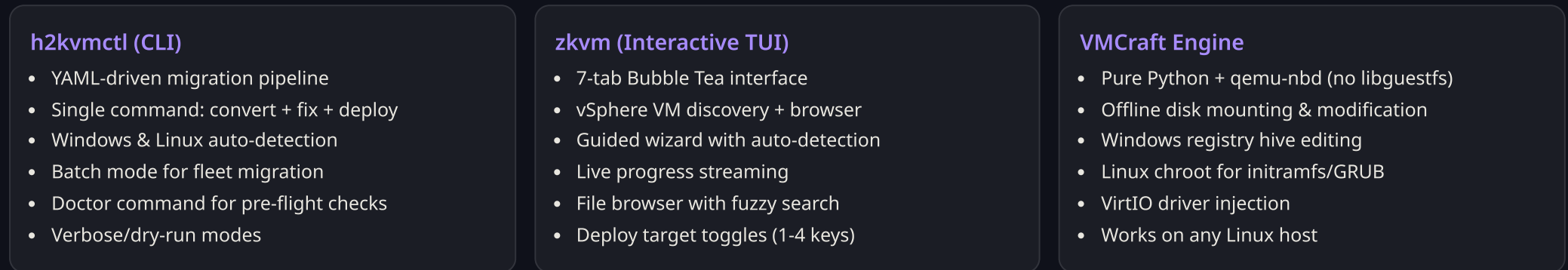
hyper2kvm is a complete VM migration toolkit: convert, fix, deploy, and verify — Windows and Linux, any source, any target.



Architecture



Core Components



hypersdk (Cloud Export Daemon)

Go daemon with hexagonal architecture supporting **10 cloud providers**: AWS, Azure, GCP, VMware vSphere, Hyper-V, Oracle Cloud, IBM Cloud, Alibaba Cloud, DigitalOcean, Hetzner. Exports VMs as raw disk images for hyper2kvm to convert and fix. REST API for automation.

The Boot Problem

Why VMs crash after migration — and how hyper2kvm prevents every failure for both Windows and Linux.

Windows Failures

Failure	hyper2kvm Fix
BSOD 0x7B — boot disk invisible	Injects viostor/vioscsi from virtio-win ISO
No network — VMXNET3/netvsc gone	Injects NetKVM VirtIO network driver
VMware Tools crash	Removes services, drivers, tray applet
Hyper-V IC conflict	Removes IC drivers from registry
BCD boot config wrong	Updates BCD for VirtIO controller
Wrong HAL — ACPI mismatch	Configures HAL for KVM (q35)
AWS/Azure agent errors	Disables cloud agent services
Memory pressure	Injects VirtIO Balloon driver
No serial console	Enables COM1 for headless access

Linux Failures

Failure	hyper2kvm Fix
Kernel panic — VFS mount fail	Rebuilds initramfs with VirtIO modules
Root mount fail — /dev/sda gone	Converts fstab to UUID-based mounts
GRUB rescue — wrong root=	Regenerates grub.cfg with correct device
No network — vmxnet3 config gone	Removes old NIC configs, loads virtio_net
SELinux denial storm	Triggers /.autorelabel on first boot
open-vm-tools errors	Removes package and service
Hyper-V daemons crash	Disables hv_kvp/vss/fcopy daemons
cloud-init re-provision	Disables cloud-init reprovisioning
LVM not found	Includes dm-mod + LVM in initramfs

All fixes applied offline — hyper2kvm mounts the disk via qemu-nbd, modifies the Windows registry / Linux filesystem directly, then unmounts. VMs never boot in a broken state. **20+ failure modes eliminated automatically.**

Supported Operating Systems

Every modern Windows and Linux version — tested and verified.

Windows

OS	Editions	Boot Mode	Status	Notes
Windows Server 2025	Standard, Datacenter	UEFI, BIOS	Tested	Latest VirtIO, Q35 recommended
Windows Server 2022	Standard, Datacenter, Core	UEFI, BIOS	Tested	Primary enterprise target
Windows Server 2019	Standard, Datacenter	UEFI, BIOS	Tested	Most common in VMware estates
Windows Server 2016	Standard, Datacenter	UEFI, BIOS	Tested	Extended support until 2027
Windows 11	Pro, Enterprise	UEFI + TPM	Tested	TPM 2.0 via swtpm on KVM
Windows 10	Pro, Enterprise, LTSC	UEFI, BIOS	Tested	VDI and workstation migrations
Windows Server 2012 R2	Standard, Datacenter	BIOS	Tested	Legacy — older VirtIO driver set

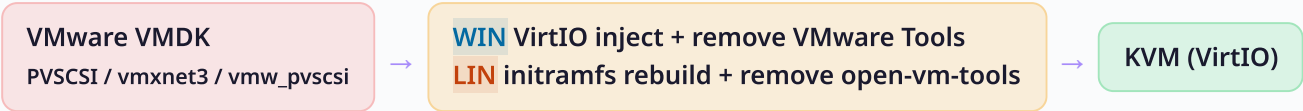
Linux

Distribution	Versions	Boot	Filesystems	Status	initramfs	Notes
RHEL	7, 8, 9, 10	UEFI/BIOS	XFS, ext4, LVM	Tested	dracut	Enterprise #1 target
Rocky / Alma	8, 9	UEFI/BIOS	XFS, ext4, LVM	Tested	dracut	CentOS replacements
Ubuntu	20.04, 22.04, 24.04	UEFI/BIOS	ext4, ZFS, LVM	Tested	update-initramfs	Cloud & server standard
Debian	11, 12	UEFI/BIOS	ext4, XFS, LVM	Tested	update-initramfs	Stable server base
SUSE SLES	12 SP5, 15 SP5/SP6	UEFI/BIOS	Btrfs, XFS, ext4	Tested	dracut	SAP & enterprise
Fedora	38-41	UEFI/BIOS	XFS, Btrfs, ext4	Tested	dracut	Cutting-edge kernel
Oracle Linux	7, 8, 9	UEFI/BIOS	XFS, ext4, LVM	Tested	dracut	UEK + RHCK support
Photon OS	3, 4, 5	UEFI/BIOS	ext4	Tested	dracut	VMware appliance OS
Amazon Linux	2, 2023	UEFI/BIOS	XFS, ext4	Tested	dracut	AWS repatriation

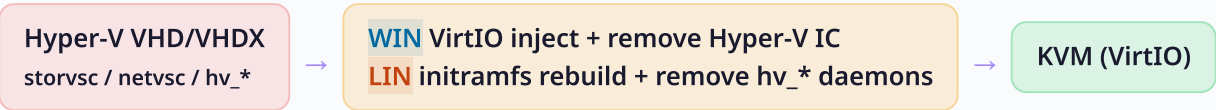
Every Source Platform

Migrate from any hypervisor or cloud — each with source-specific fixes for both Windows and Linux.

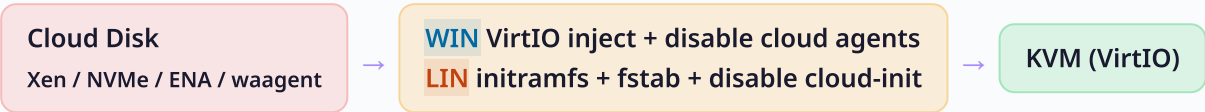
VMware vSphere → KVM



Hyper-V → KVM



AWS / Azure / GCP → KVM



Unified Fix Matrix — All Sources × Both OS

Fix	OS	VMware	Hyper-V	AWS	Azure	GCP	OpenStack
VirtIO storage driver	W	✓	✓	✓	✓	✓	✓
initramfs + VirtIO	L	✓	✓	✓	✓	✓	✓
VirtIO network driver	W	✓	✓	✓	✓	✓	✓
fstab UUID repair	L	✓	✓	✓	✓	✓	✓
BCD / GRUB repair	WL	✓	✓	✓	✓	✓	✓
Remove VMware Tools	WL	✓	N/A	N/A	N/A	N/A	N/A
Remove Hyper-V IC	WL	N/A	✓	N/A	✓	N/A	N/A
Remove cloud agents	WL	N/A	N/A	✓	✓	✓	N/A
SELinux relabel	L	✓	✓	✓	✓	✓	✓

Serial console

WL



Windows Migration Deep Dive

The complete Windows offline fix pipeline — every stage explained.

Stage 1: Bootstrap

- Mount disk offline via qemu-nbd
- Detect Windows version (2016-2025, 10/11)
- Detect boot mode (BIOS vs UEFI)
- Detect storage controller (LSI/PVSCSI/storvsc)
- Locate registry hives (SYSTEM, SOFTWARE)
- Map partition layout (EFI, Recovery, C:)
- Select VirtIO driver version for OS
- Identify installed hypervisor tools

Stage 2: Driver Injection

- Extract drivers from virtio-win ISO
- Copy viostor/vioscsi .sys + .inf to C:\Windows
- Copy NetKVM (network), Balloon (memory)
- Load offline SYSTEM hive
- Add registry entries (Start=0, boot-start)
- Update CriticalDeviceDatabase for VirtIO PCI
- Configure UpperFilters/LowerFilters
- Set correct driver stack ordering

Stage 3: Cleanup & Verify

- Remove VMware/Hyper-V/cloud services
- Remove source hypervisor drivers
- Update BCD for new boot controller
- Enable serial console (COM1)
- Generate libvirt XML (Q35 + VirtIO)
- Generate KubeVirt manifest (optional)
- Automated boot test (no BSOD check)
- Webhook notification on result

Windows YAML Configuration

```
# windows-migration.yaml
cmd: vsphere                # Or: local, hyperv
vcenter: vcenter.corp.com
vs_action: export_vm
vs_vm: win-sql-server-01
output_dir: ./converted
out_format: qcow2
windows: true                # Enable Windows pipeline
guest_os: windows
virtio_win_iso: /usr/share/virtio-win/virtio-win.iso
emit_domain_xml: true        # libvirt XML
deploy_k8s: false            # KubeVirt manifest
vm_name: win-sql-server-01
memory: 8192
vcpus: 4
machine: q35                 # PCIe chipset
uefi: true                   # UEFI boot
boot_test: true              # Auto boot verification
```

VirtIO Driver Components Injected

Driver	File	Purpose	Registry Configuration
viostor	viostor.sys	VirtIO block storage (boot-critical)	Start=0, Group=SCSI miniport, CriticalDeviceDatabase

vioscsi	vioscsi.sys	VirtIO SCSI controller	Start=0, Group=SCSI miniport
NetKVM	netkvm.sys	VirtIO network adapter	Start=3, NDIS miniport registration
Balloon	balloon.sys	Dynamic memory management	Start=3, blnsvr service
vioserial	vioser.sys	Serial port for QEMU guest agent	Start=3, serial device class
viornrg	viornrg.sys	Hardware RNG from KVM host	Start=3, entropy provider

Linux Migration Deep Dive

The complete Linux offline fix pipeline — every fix explained.



#	Fix	What It Does	Technical Detail
1	OS Detection	Identifies distro, version, init, boot mode, partitions	Parses /etc/os-release, EFI presence, GPT/MBR, LVM VGs
2	fstab UUID Repair	Converts /dev/sdX mounts to UUID-based	blkid scan, replaces /dev/sdX, /dev/xvdX, /dev/hdX with UUID=
3	initramfs Rebuild	Injects virtio_blk, virtio_scsi, virtio_net, virtio_pci	Chroot: dracut (RHEL/Fedora/SUSE) or update-initramfs (Debian/Ubuntu)
4	GRUB Update	Fixes root= kernel parameter	Updates GRUB_CMDLINE_LINUX, regenerates grub.cfg via grub2-mkconfig
5	Network Cleanup	Removes old NIC configs (vmxnet3/hv_netvsc)	Deletes ifcfg-*, netplan YAML with hardware MACs, 70-persistent-net.rules
6	VMware Tools	Removes open-vm-tools	rpm -e / apt remove, disable vmtoolsd.service
7	Hyper-V IC	Removes Hyper-V daemons	Disable hv_kvp_daemon, hv_vss_daemon, hv_fccopy_daemon
8	Cloud Agents	Disables AWS/Azure/GCP agents	Disable cloud-init, waagent, SSM agent; preserve hostname
9	SELinux Relabel	Triggers filesystem relabel	Creates /.autorelabel for restorecon on first boot
10	Serial Console	Enables ttyS0 for headless access	console=ttyS0,115200n8 in GRUB, enable serial-getty@ttyS0

Linux YAML Configuration

```
# rhel9-webserver.yaml
cmd: vsphere
vcenter: vcenter.corp.com
vs_action: export_vm
vs_vm: rhel9-web-01
output_dir: ./converted
out_format: qcow2
regen_initramfs: true
emit_domain_xml: true
vm_name: rhel9-web-01
memory: 4096
vcpus: 2
machine: q35
uefi: true

# aws-postgres.yaml (KubeVirt target)
cmd: local
raw: ./aws-export/postgres.raw
output_dir: ./converted
out_format: qcow2
regen_initramfs: true
deploy_k8s: true
k8s_namespace: databases
vm_name: postgres-primary
memory: 16384
vcpus: 8
boot_test: true
```

boot_test: true

Enterprise Workloads — Windows

Critical Windows applications migrated safely with tested procedures.

SQL Server

- SQL 2016, 2017, 2019, 2022 — all tested
- Database files migrate with disk intact
- VirtIO SCSI matches PVSCSI performance
- TempDB, log files, data files preserved
- KVM vCPU pinning cuts per-core license 20-40%
- Always-On AG: migrate secondaries first

Active Directory

- NTDS.dit database migrates with disk
- SYSVOL/NETLOGON shares preserved
- AD replication resumes after KVM boot
- Strategy: secondaries first, verify, then primary
- DNS zones and records intact
- Optional MAC address preservation

Exchange Server

- Exchange 2016, 2019 tested
- Mailbox databases mount post-migration
- DAG: migrate passive copies first
- Certificate stores preserved in registry
- Transport queues: drain before migration
- Put in maintenance mode during cutover

IIS / .NET Web Apps

- Sites, app pools, bindings preserved
- SSL certificates in cert store intact
- .NET applications run unchanged
- Simplest migration — standard VirtIO injection
- Load balancer health checks resume
- Verify URL bindings post-migration

File Servers (NTFS/ReFS)

- NTFS permissions (ACLs) fully preserved
- DFS namespaces and replication intact
- SMB shares auto-start on boot
- Deduplication volumes migrate correctly
- Shadow copies preserved with disk
- FSRM quotas maintained

Remote Desktop Services

- Session host, broker, gateway roles migrate
- User profiles and settings preserved
- RDP certificates intact
- GPU passthrough for RemoteFX on KVM
- Connection broker DB migrates with SQL
- Licensing server: reactivate post-migration

SQL Server Licensing Savings with KVM

VMware Problem

VMware reports all physical cores to Microsoft. SQL per-core licensing counts host cores. A 2-socket 16-core host = 32 cores to license. Even 4-vCPU SQL VM may require all 32 cores licensed. **SQL Enterprise: \$15,123/core = \$483,936/host.**

KVM Advantage

KVM with CPU pinning exposes only assigned cores. 4 pinned vCPUs = license only 4 cores. NUMA-aware placement, cgroups enforce isolation (auditor-verifiable). **SQL Enterprise: \$15,123 x 4 = \$60,492. Savings: \$423K per host.**

Enterprise Workloads — Linux

Critical Linux applications migrated safely with tested procedures.

PostgreSQL / MySQL / MariaDB

- Data directories migrate with disk
- InnoDB/WAL files intact on target
- VirtIO SCSI comparable to PVSCSI I/O
- Tablespaces on separate partitions via fstab fix
- Replication resumes (update source IP)
- Stop DB before export for consistency

Docker / Podman Hosts

- Container images in /var/lib/docker intact
- Overlay2/fuse-overlayfs storage preserved
- Docker daemon starts, containers resume
- Podman rootless containers preserved
- Registry data (Harbor, GitLab) travels with vol
- docker-compose stacks auto-restart via systemd

Kubernetes Nodes

- kubelet, containerd, etcd data migrates
- Migrate workers first, then control plane
- etcd snapshot before control plane migration
- CNI (Calico, Cilium) needs network reconfig
- PersistentVolumes on local disk travel with VM
- Cordon + drain node before migration

Web Servers (Apache / Nginx)

- Simplest migration — standard pipeline
- Virtual hosts, SSL certs, .htaccess preserved
- Let's Encrypt certs intact
- PHP-FPM, Python WSGI, Node.js unchanged
- Load balancer health checks resume
- Verify DNS/IP bindings post-migration

NFS / Samba File Servers

- Exported filesystems migrate with disk
- NFS /etc/exports preserved
- Samba smb.conf intact
- POSIX ACLs + xattrs on ext4/XFS
- Clients reconnect after DNS/IP update
- Update exports for new network if IP changes

Monitoring / CI-CD

- Prometheus TSDB, Grafana dashboards intact
- Elasticsearch indices travel with disk
- GitLab PostgreSQL + Redis data preserved
- Jenkins workspace/job history intact
- SSH keys in /home/*/.ssh preserved
- VirtIO SCSI + NVMe for heavy I/O

FreeIPA / LDAP / Kerberos

- 389-ds LDAP directory migrates with disk
- Kerberos keytabs/principal DB preserved
- CA certificates in /etc/pki intact
- DNS zones travel with LDAP
- Replicas first, verify sync, then primary
- Test kinit + ldapsearch post-migration

SAP / Oracle DB on SUSE/RHEL

- SAP HANA: memory-mapped files travel with disk
- Oracle DB datafiles, redo logs intact
- Huge pages configured in KVM XML
- CPU pinning for consistent performance
- SUSE SLES certified for KVM
- RHEL certified for KVM virtualization

Deployment Targets

Deploy converted VMs to KVM/libvirt, KubeVirt/Kubernetes, or both simultaneously.

KVM / libvirt

- `emit_domain_xml: true`
- Optimized libvirt domain XML generation
- Q35 machine type with PCIe
- VirtIO disk, network, balloon, RNG, serial
- UEFI (OVMF) or BIOS boot
- TPM 2.0 via swtpm (Windows 11)
- CPU pinning, NUMA, huge pages
- Auto virsh define + optional boot test
- VNC/Spice console for visual verification

KubeVirt / Kubernetes

- `deploy_k8s: true`
- VirtualMachine manifest generation
- PVC with disk image upload via CDI
- Works on K3s, vanilla K8s, OpenShift
- GitOps-ready YAML for fleet deploy
- kubectl apply to any cluster
- Resource requests/limits from VM config
- Network policies, pod affinity
- Live migration support on KubeVirt

Dual Target Deploy

- Both outputs from one conversion run
- `emit_domain_xml: true` + `deploy_k8s: true`
- KVM for traditional bare-metal hosts
- KubeVirt for Kubernetes clusters
- Choose per-site at deploy time
- Same QCOW2 image for both targets
- Edge sites: K3s + KubeVirt
- Data center: libvirt + KVM
- Hybrid: mix and match per workload

libvirt XML Features Generated

Feature	Configuration	Benefit
Machine type	Q35 (PCIe) or i440FX (legacy)	Modern chipset with PCIe for better device passthrough
CPU model	host-passthrough with pinning	Full host CPU features, NUMA-aware, license compliance
Disk	VirtIO SCSI with write-back cache	Near-native disk I/O performance
Network	VirtIO-net with multiqueue	10 Gbps+ throughput with RSS
Firmware	OVMF (UEFI) or SeaBIOS	Secure Boot, measured boot chain
TPM	swtpm (TPM 2.0 emulation)	Windows 11, BitLocker, attestation
Console	VNC + serial (ttyS0/COM1)	Graphical + headless management
Memory	Balloon + huge pages	Dynamic memory + deterministic performance
RNG	VirtIO RNG from /dev/urandom	Fast entropy for crypto operations
GPU	VFIO passthrough support	GPU workloads, VDI, ML/AI

zkvm Interactive TUI

A guided, 7-tab terminal interface for interactive migration — no YAML editing required.

7 Tabs

- **Home** — welcome, quick start guide
- **Dashboard** — system status, recent migrations
- **Wizard** — guided step-by-step migration form
- **Browser** — vSphere VM discovery + selection
- **Migrations** — active/completed migration tracking
- **Batch** — multi-VM queue management
- **Settings** — configuration preferences

Wizard Features

- **Source auto-detection:** .vmdk/.ova/.vhd/.raw
- **File browser:** fuzzy search, extension filter
- **Category navigation:** j/k keys, Enter to expand
- **Deploy toggles:** keys 1-4 for targets
- **Live execution plan:** shows what will happen
- **Progress streaming:** real-time with ETA

vSphere Discovery

- Connects to vCenter via govc
- Browses datacenter/cluster/host hierarchy
- Shows VM details: OS, CPU, RAM, disk, power state
- Multi-select VMs for batch migration
- Ctrl+R syncs selection to wizard form
- Auto-sets cmd=vsphere, vs_action=export_vm
- Environment variables: GOVC_URL, _USERNAME, _PASSWORD

Progress & Monitoring

- Streaming progress with \r handling
- govc download: throttled (1 per 3s, dedup)
- File-size progress with total from vm.info
- [PROGRESS] lines: pct|current|total|rate|eta
- 5-sample moving average for speed smoothing
- Progress bar in right panel
- Auto-sudo when not root

CLI Tools

h2kvmctl (CLI)

```
# Convert + fix + deploy
h2kvmctl -c migration.yaml

# Doctor (pre-flight checks)
h2kvmctl doctor

# Dry run
h2kvmctl -c config.yaml --dry-run
```

doctor.sh (System Readiness)

- 9 system areas checked with timeouts
- qemu-nbd, qemu-img availability
- virtio-win ISO presence (Windows)
- libvirt/virsh installation
- Disk space verification
- KVM kernel module loaded
- Network bridge configuration
- KubeVirt/kubectrl (if K8s target)

Automation Scripts

- **quickstart.sh** — fresh machine setup
- **run-demo.sh** — convert + boot + SSH
- **record-demo.sh** — asciinema recording
- Webhook integration (Slack/Teams)
- CI/CD pipeline integration
- Ansible/Terraform compatible

```
# Verbose output  
h2kvmctl -c config.yaml -v
```

- Color-coded pass/fail output

Boot Verification & Safety

Every migration — Windows and Linux — verified before production cutover.

Automated Boot Test

After conversion, hyper2kvm starts the VM on KVM, monitors boot progress, and confirms it reaches login (Windows) or shell (Linux). **No human needs to watch.** Fails are reported immediately with the error.

VNC/noVNC Console

Full graphical console for visual verification. See Windows desktop or Linux login prompt. Debug issues in real-time. Browser-based via noVNC — no client needed.

Post-Boot Health Checks

- Disk I/O: VirtIO SCSI responding
- Network: IP assigned, ports open
- WIN RDP accessible on 3389
- LIN SSH accessible on 22
- Critical services running (SQL, httpd, etc.)
- Event log / journal scan for errors
- Serial console functional

Zero-Risk Rollback

- Source disk preserved until verification
- Original VM untouched on source platform
- Conversion creates a copy — never modifies source
- If boot fails: fix and retry, source is safe
- Webhook alerts on success/failure
- Auditable pass/fail report per VM

8-Point Validation Checklist

#	Check	OS	What It Validates	Why It Matters
1	Pre-flight (doctor)	WL	Disk space, tools, ISO, libvirt	Don't start if dependencies missing
2	Source disk integrity	WL	VMDK descriptor, partition table	Catch corrupt disks early
3	OS version detect	WL	Registry / os-release	Select correct fix strategy
4	Boot mode detect	WL	EFI System Partition present?	UEFI vs BIOS affects config
5	Driver/initramfs verify	WL	viostor.sys / VirtIO in initramfs	Confirm driver loads at boot
6	Source tools removed	WL	VMware/Hyper-V services disabled	Prevent conflicts
7	Boot test	WL	VM reaches login/shell, no BSOD/panic	Ultimate validation
8	Network test	WL	IP assigned, RDP/SSH port open	VM reachable after cutover

Tool Comparison

hyper2kvm vs. every alternative — for both Windows and Linux migration.

Capability	hyper2kvm	virt-v2v	Red Hat MTV	AWS MGN	Azure Migrate	Manual
Windows VirtIO injection	✓ Auto	✓	✓	N/A	N/A	Manual
Linux initramfs rebuild	✓ All distros	✓	Limited	N/A	N/A	Manual
Windows registry fix	✓ Auto	✓	Basic	N/A	N/A	Manual
Linux fstab/GRUB fix	✓ Auto	✓	Basic	N/A	N/A	Manual
Cloud agent removal	✓ AWS/Azure/GCP	✗	✗	N/A	N/A	Manual
Boot verification	✓ Auto	✗	✗	✗	✗	Manual
Deploy to KVM	✓	✓	✗	✗	✗	Manual
Deploy to KubeVirt	✓	✗	✓ (OCP)	✗	✗	✗
Interactive TUI	✓ zkvm	✗	Web UI	✗	Web UI	✗
Batch migration	✓	✗	✓	✓	✓	✗
Source: VMware	✓	✓	✓	✓	✓	Manual
Source: Hyper-V	✓	✓	Partial	✓	✓	Manual
Source: AWS/Azure/GCP	✓ (10 clouds)	✗	✗	AWS only	Azure only	Complex
No libguestfs	✓ VMCraft	Required	Required	N/A	N/A	N/A
15+ Linux distros	✓	✓	RHEL-focus	Amazon	Ubuntu	Varies
TPM emulation (Win11)	✓ swtpm	✗	✓	N/A	N/A	Manual
Cost	\$0	\$0	\$50K+ (OCP)	Free/90d	Free/180d	\$0

The Only Tool That Does It All

hyper2kvm is the only tool that migrates **both Windows & Linux** from **VMware + Hyper-V + 10 clouds**, deploys to **KVM + KubeVirt**, includes **automated boot verification**, needs **no libguestfs**, has an **interactive TUI**, and costs **\$0**.

Batch Migration Workflow

Migrate 10 or 1,000 VMs with the same pipeline — mixed Windows and Linux.

End-to-End Batch Pipeline

#	Step	Tool	What Happens	Output
1	Pre-flight	h2kvmctl	h2kvmctl doctor — verifies disk space, tools, ISOs, libvirt, KubeVirt	Green light to proceed
2	Discover	zkvm TUI	Browse vCenter: see OS, CPU, RAM, disk for every VM	VM selection list
3	Export	govc / hypersdk	Streaming download with progress + ETA (3s throttled updates)	VMDK/VHD/RAW files
4	Convert	qemu-img	VMDK/VHD → QCOW2, flatten snapshots, compress	Optimized QCOW2 images
5	Detect	VMCraft	Auto-detect: Windows or Linux? Version? Boot mode? Filesystem?	OS-specific fix strategy
6	Fix (Windows)	VMCraft	VirtIO inject, registry fix, BCD repair, tools removal	Boot-ready Windows VM
7	Fix (Linux)	VMCraft	initramfs rebuild, fstab, GRUB, network, SELinux	Boot-ready Linux VM
8	Deploy	h2kvmctl	Generate XML + virsh define, or K8s YAML + kubectl apply	VMs defined on targets
9	Verify	h2kvmctl	Automated boot test per VM, health checks	Pass/fail report
10	Notify	Webhook	Slack/Teams notification with per-VM results	Team informed

Real-World Scale Examples

VMware Exit: 250 VMs (Mixed OS)

- 150 Linux (RHEL 8/9, Ubuntu 22.04) + 100 Windows (2019/2022)
- Source: vSphere 7, 8 ESXi hosts
- Target: 12 RHEL 9 KVM hosts
- Timeline: 6 weeks (40+ VMs/week, off-hours cutover)
- Savings: \$500K/yr VMware + \$632K SQL licensing = **\$1.13M/yr**

Multi-Cloud Consolidation: 100 VMs

- 30 AWS + 25 Azure + 15 GCP + 30 VMware — mixed Win/Linux
- Target: centralized KVM cluster + edge K3s (KubeVirt)
- hypersdk exports from all 4 sources in parallel
- hyper2kvm applies source-specific + OS-specific fixes
- Dual-target deploy: libvirt + KubeVirt per workload
- Savings: **\$1.5M/yr** consolidated infrastructure

Hyper-V to KubeVirt: 60 VMs

- 20 Windows Server + 40 Linux (Debian, Ubuntu, RHEL)
- Source: Hyper-V 2019 cluster
- Target: K3s + KubeVirt (modernization)
- KubeVirt manifests generated per VM
- kubectl apply — entire fleet running on K8s
- Savings: \$200K/yr Hyper-V + Windows Datacenter

Azure Repatriation: 40 VMs

- 15 Windows (.NET + SQL) + 25 Linux (PostgreSQL, Django)
- Source: Azure D/E-series VMs
- Motivation: data sovereignty + cost control
- hypersdk exports managed disks
- hyper2kvm removes Azure agent + Hyper-V IC
- Savings: **\$720K/yr** in Azure compute + egress

Cost Savings & ROI

Eliminate hypervisor licensing, reduce cloud spend, cut SQL Server costs — hyper2kvm pays for itself immediately.

\$0

hyper2kvm license
(Apache 2.0)

\$0

KVM hypervisor
license cost

20-40%

SQL Server savings
with CPU pinning

50-80%

Cloud cost savings
on repatriation

3-Year TCO Comparison: 100 VM Estate (Mixed Win/Linux)

Cost Item	VMware vSphere	AWS/Azure Cloud	KVM (hyper2kvm)	Best Savings
Hypervisor licensing	\$450,000	N/A	\$0	\$450,000
Compute costs	\$180,000 (hardware)	\$2,400,000	\$180,000 (hardware)	\$2,220,000
SQL Server (10 VMs)	\$1,263,600	\$1,263,600	\$631,200 (pinned)	\$632,400
Windows Server	\$360,000	\$360,000	\$360,000	\$0
Migration (one-time)	N/A	N/A	\$25,000	-\$25,000
Support / operations	\$270,000	\$150,000	\$225,000	\$45,000
3-Year Total	\$2,523,600	\$4,173,600	\$1,421,200	Up to \$2.75M

VMware Exit Savings

- Eliminate vSphere/vCenter licensing
- No per-CPU socket fees
- No vSAN licensing
- No NSX licensing
- Community KVM support is free
- Typical: \$5K-15K/host/yr saved

Cloud Repatriation Savings

- Eliminate compute instance fees
- Eliminate egress charges
- Eliminate managed service premiums
- One-time hardware investment
- 3-year payback typical
- Typical: 50-80% TCO reduction

SQL Server + KVM Savings

- CPU pinning reduces licensed cores
- NUMA-aware placement
- cgroups isolation (audit-verifiable)
- SQL Standard: \$3,945/core saved
- SQL Enterprise: \$15,123/core saved
- Typical: \$100K-500K/yr per estate

Security, Compliance & Advanced

Enterprise-grade security features for regulated environments.

Security Features

- Offline processing — no network during fix
- Source disks never modified
- UEFI Secure Boot support
- TPM 2.0 emulation (swtpm)
- BitLocker-aware (Windows)
- LUKS encryption support (Linux)
- SELinux autorelabel (RHEL/Fedora)
- No cloud metadata exposure

Compliance

- Auditable migration reports
- Checksum verification (SHA256)
- Air-gap compatible (no internet needed)
- FedRAMP / HIPAA / SOC2 environments
- Data sovereignty (on-prem target)
- Change management audit trail
- CPU pinning for license compliance
- Apache 2.0 — no vendor lock-in

Advanced Features

- Snapshot flattening on convert
- QCOW2 compression (qcow2+zlib)
- Multi-disk VM support
- GPU passthrough (VFIO)
- SR-IOV network device passthrough
- NVMe backend for storage
- Huge pages for deterministic perf
- Live migration ready on KubeVirt

Filesystem & Volume Manager Support

Windows Filesystems

- **NTFS** — full ACL preservation, dedup support
- **ReFS** — Storage Spaces Direct volumes
- **FAT32 (EFI)** — EFI System Partition handling
- **GPT + MBR** — both partition schemes

Linux Filesystems & Volumes

- **ext4, XFS, Btrfs, ZFS** — all major filesystems
- **LVM2** — VG/LV detection, module injection
- **mdadm RAID** — software RAID arrays
- **LUKS** — encrypted root support
- **GRUB2, GRUB Legacy, systemd-boot**

Network Architecture

Feature	KVM / libvirt	KubeVirt / K8s
Default NIC	VirtIO-net (virtio_net)	Multus + bridge / macvlan
Performance	Multiqueue VirtIO-net, 10G+	SR-IOV passthrough available
Bridge	Linux bridge or Open vSwitch	CNI bridge plugin
VLAN	802.1Q trunk to VMs	Multus secondary networks
Bonding	bond0 on host, bridge on top	K8s node bonding

Firewall

nftables/iptables per host

NetworkPolicy per namespace

Complete Feature Summary

Everything hyper2kvm does — in one view.

Source Platforms (10)

- VMware vSphere 6.5-8.0
- Microsoft Hyper-V 2016-2022
- AWS EC2
- Microsoft Azure
- Google Cloud Platform
- Oracle Cloud Infrastructure
- IBM Cloud
- Alibaba Cloud
- DigitalOcean
- Hetzner Cloud

Operating Systems (22+)

- Windows Server 2012 R2 – 2025
- Windows 10, Windows 11
- RHEL 7-10, Rocky 8-9, Alma 8-9
- Ubuntu 20.04-24.04
- Debian 11-12
- SUSE SLES 12-15, openSUSE Leap/TW
- Fedora 38-41
- Oracle Linux 7-9
- Photon OS 3-5, Amazon Linux 2/2023
- CentOS Stream 8-9, CentOS 7

Automatic Fixes (20+)

- **W** VirtIO driver injection (6 drivers)
- **W** Registry driver stack configuration
- **W** BCD boot repair
- **W** HAL configuration
- **L** initramfs rebuild (all distros)
- **L** fstab UUID repair
- **L** GRUB config regeneration
- **L** SELinux autorelabel
- VMware/Hyper-V/cloud tools removal
- Network cleanup + serial console

Deployment & Tooling

- KVM / libvirt (XML generation)
- KubeVirt / K8s (YAML manifests)
- Dual-target deploy
- h2kvmctl CLI + YAML config
- zkvm interactive TUI (7 tabs)
- hypersdk cloud export daemon
- Automated boot verification
- Batch migration (100s of VMs)
- Webhook notifications
- CI/CD + Ansible integration

Migrate Everything. From Anywhere. To KVM.

Windows + **Linux** | 10 Source Platforms | 22+ OS Versions | 20+ Automatic Fixes

KVM + KubeVirt | Interactive TUI + CLI + API | Automated Boot Verification

No libguestfs | No vendor lock-in | **\$0 license cost** | **Apache 2.0 Open Source**

github.com/ssahani/hyper2kvm

Source Code

[h2kvmctl -c config.yaml](#)

Get Started

[zkvm](#)

Interactive TUI

hyper2kvm + hypersdk — Enterprise VM Migration Platform | Version 0.2.2 | Apache 2.0 Open Source